Instruction Tuning and Direct Preference Optimization on Small Language Models

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1 Repository, models, and data

Repository: https://github.com/AhmadAkhtar007/pa2-small-model-alignment. The main experiments start from `Qwen/Qwen2.5-0.5B`; the persona experiment uses `HuggingFaceTB/SmolLM2-360M-Instruct`. The repository contains the custom SFT and DPO loops, the root submission adapter, tests, configurations, fixed evaluation sets, and scripts for running each stage.

Track	Training	Validation	Evaluation
SFT / UltraChat	30,000	1,000	Shared HH and AlpacaEval sets
HH DPO / HH-RLHF	15,000	1,000	300 supplied synthetic HH prompts
UF DPO / UltraFeedback	15,000	1,000	300 fixed AlpacaEval prompts
Persona LoRA	198	22	12 supplied persona prompts

For UltraChat, each example uses the first user message and assistant response. HH and UltraFeedback pairs are restricted to single-turn conversations. Duplicate prompts are removed before sampling and splitting, and checks confirmed that each main track has unique prompts with no train/validation overlap. All random splits and the fixed AlpacaEval subset use seed 42. The persona experiment splits the supplied 220 preferences into 198 training and 22 validation pairs. Saved training outputs retain the prepared-data fingerprints and SHA256 hashes; the repository includes the evaluation files and generation metadata.

2 Experimental configuration

Setting	SFT	HH DPO	UF DPO	Persona LoRA
Trainable strategy	Full model	Full model	Full model	LoRA
Optimizer	AdamW	AdamW	AdamW	AdamW
Peak learning rate	2×10^{-5}	10^{-6}	10^{-6}	5×10^{-5}
Schedule / warmup	Cosine / 3%	Constant / 0	Constant / 0	Cosine / 3%
Weight decay	0.1	0	0	0
Microbatch / accumulation	2 / 16	1 / 32	1 / 32	1 / 8
Effective batch	32	32	32	8
Configured epochs	1	1	1	3
Successful updates	938	465	465	72
Sequence / prompt cap	512 / 256	512 / 256	512 / 256	512 / 256
DPO β	—	0.1	0.1	0.1
Reference	—	SFT best	SFT best	Frozen instruct base

All runs use gradient clipping at 1.0. The custom loops keep trainable weights in FP32 and use FP16 autocast, gradient scaling, and gradient checkpointing. On Kaggle T4 ×2, each DPO run puts the policy on GPU 0 and the frozen FP16 reference on GPU 1. HH and UF ran sequentially, each starting independently from the same best SFT checkpoint. The persona run uses TRL 0.15.2 and PEFT 0.14.0. Its LoRA settings are rank 8, alpha 16, dropout 0, no adapted bias, and target modules `q_proj`, `k_proj`, `v_proj`, and `o_proj`. Disabling the trainable adapter gives the frozen base-model reference.

3 Implementation and parameter choices

The loss scores only response tokens: prompt and padding tokens are masked out, and EOS is appended before truncation. SFT divides the total response-token negative log likelihood by the number of scored tokens in each accumulation group, including the final partial group. DPO sums response log probabilities $s_\theta(x, y)$ and averages the following objective over pairs:

$$-\log \sigma(\beta\,[s_\theta(x, y_w) - s_\theta(x, y_l) - s_{\text{ref}}(x, y_w) + s_{\text{ref}}(x, y_l)])\,.$$

The reference model runs without gradients. A validation pair counts as correct only when the chosen response has a strictly higher implicit reward than the rejected response. SFT checkpoint selection uses the lowest validation token loss; DPO selection uses the highest validation preference accuracy. These objectives and training loops are implemented in PyTorch. Only the persona experiment uses TRL's DPOTrainer.

Qwen uses the template `### Instruction:\n{instruction}\n\n### Response:\n`, where `\n` denotes a newline. SFT and evaluation both retain the last 256 tokens of the formatted prompt, which can discard part of a long instruction. SmolLM2 uses its instruct chat template. The main DPO runs follow the guide's AdamW option, learning rate, beta, and effective batch size.

For persona tuning, rank 8 keeps the adapter small. Adapting the attention query, key, value, and output projections allows style changes while the base weights remain frozen. The learning rate of 5×10^{-5} and three configured epochs were initial settings for the 198 training pairs; no tuning sweep was run. A microbatch of 1 limits memory use, with eight-step accumulation giving an effective batch of 8. The model loads in FP32 and trains with FP16 mixed precision; BF16 is disabled. Validation preference accuracy reached 100% on 22 pairs, while held-out style scores showed no improvement.

Source was transferred to Kaggle through a Google Drive ZIP and is versioned on GitHub. Base responses were generated after training from the untouched base checkpoint. The persona run completed 72 updates; its final Trainer state records epoch 2.889, with three epochs configured.

Initial greedy outputs repeated phrases, so final decoding uses a repetition penalty of 1.1 and no-repeat trigram size 3. Before generating the full sets, three responses were inspected for each of seven model/set combinations. All comparisons use the same settings on both sides: greedy decoding, one beam, seed 42, and a maximum of 256 new tokens. Temperature, top-p, and length penalty remain at 1.0.

4 Training behavior and checkpoint selection

Track	Final updates	Selected update	Best validation result
SFT	938	938	Token loss 1.373631
HH DPO	465	400	Preference accuracy 62.5%
UF DPO	465	465	Preference accuracy 68.9%
Persona LoRA	72	25	Preference accuracy 100% on 22 pairs

SFT reaches its lowest recorded validation loss at the final checkpoint. For HH, validation DPO loss falls from about 0.693 to 0.651 at update 400, then rises slightly. Preference accuracy also peaks at 400, which selects that checkpoint. UF reaches validation loss 0.585609 at its final update. Both DPO runs have increasing reward margins, so the policy separates chosen and rejected responses more strongly relative to the reference. Training losses fluctuate, and validation gains remain limited.

Both custom DPO runs process their full 15,000-pair subsets. The reported update counts exclude steps skipped by the gradient scaler because of overflow. Persona reaches 100% preference accuracy on its 22 validation pairs, but the later style evaluation remains weak. This gap limits what can be concluded from the early best checkpoint. All plots use raw saved metrics without smoothing. Dashed lines mark the selected DPO and persona checkpoints; the original full-size plots are included separately.

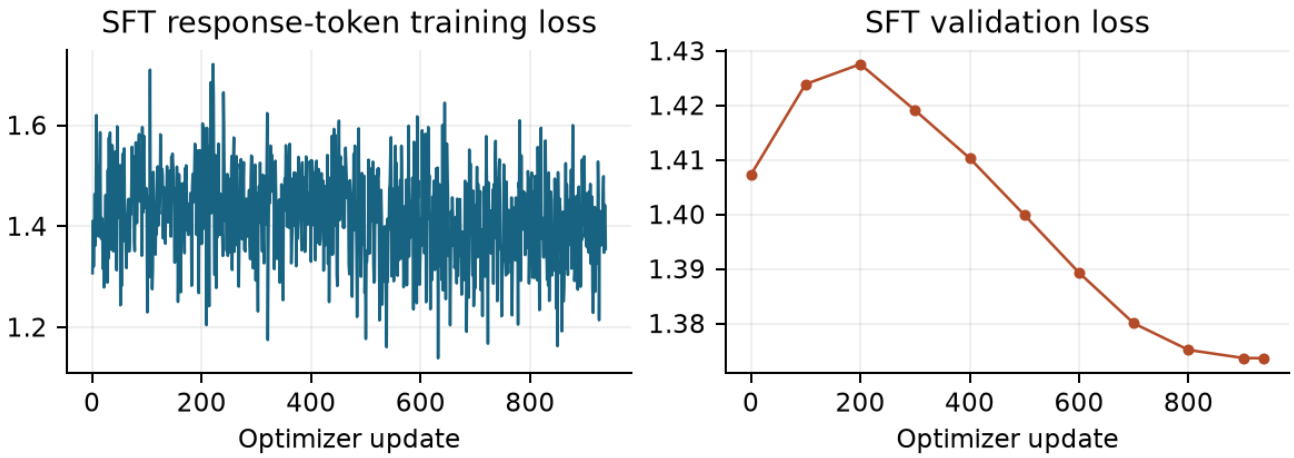


Figure 1: SFT training and validation response-token losses.



Figure 2: Persona LoRA loss and validation preference accuracy. Dashed line: selected update 25. The final Trainer state records 72 updates and epoch 2.889.

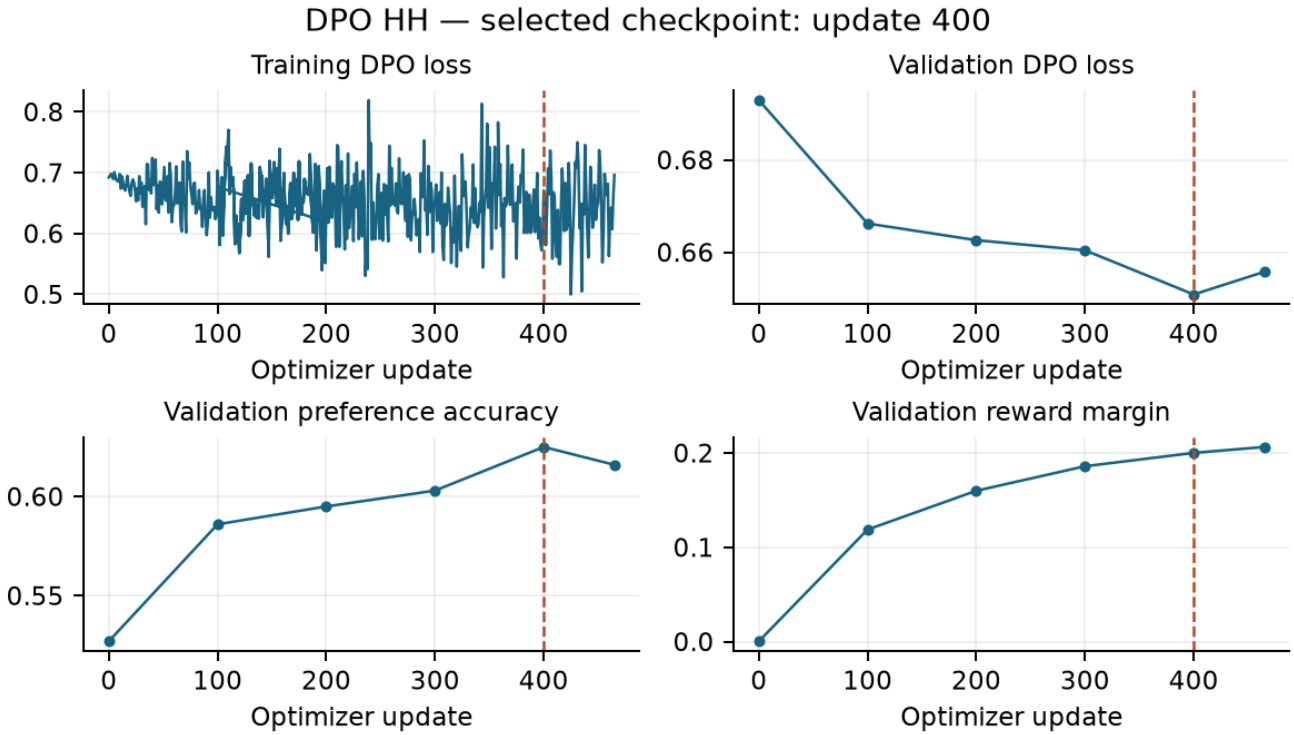


Figure 3: HH DPO diagnostics. The selected checkpoint is update 400, not the final update.

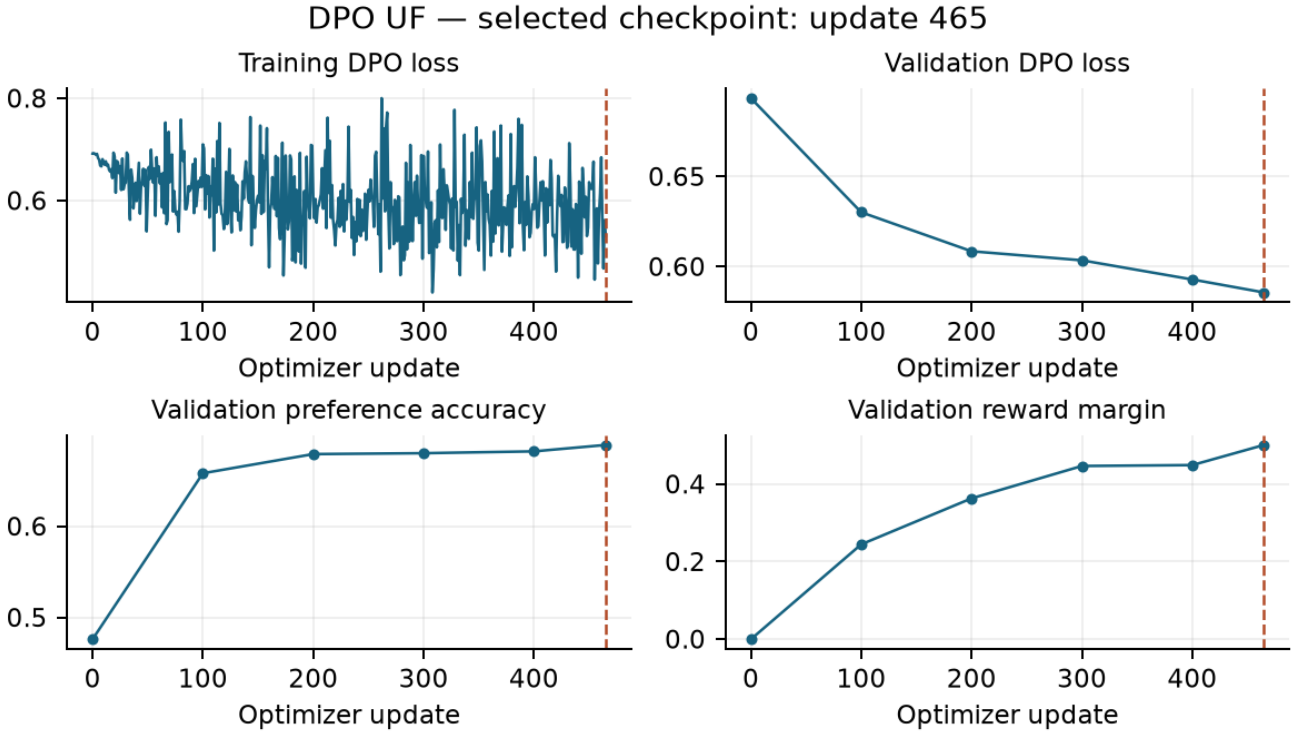


Figure 4: UF DPO diagnostics. The final update 465 is selected. Preference-validation improvement did not translate into a judged advantage over SFT.

5 DPO-versus-SFT and persona evaluation

Generation produced 1,524 nonempty responses: base/SFT/UF on AlpacaEval, SFT/HH on the supplied HH set, and base/adapted SmolLM2 on the persona set. Checks verified the response counts, unique IDs, pair mappings, and each judgment's pair hash. The judge sees only Response A and Response B, with positions randomized using seed 42. A separate Kaggle utility benchmark notebook calls `google/gemini-3.6-flash` to judge the saved pairs.

Candidate vs baseline	N	Wins	Losses	Ties	Decisive WR	Ties-half WR
HH DPO vs SFT	300	168	95	37	63.88%	62.17%
UF DPO vs SFT	300	95	138	67	40.77%	42.83%
Persona LoRA vs instruct	12	4	4	4	50.00%	50.00%

Decisive WR is $W/(W+L)$; ties-half WR is $(W+T/2)/(W+L+T)$. All three sets have zero final unknowns, unresolved failures, or pending comparisons. HH has 73 more wins than losses; UF has 43 more losses than wins. Scores are reported separately because the evaluation tasks differ.

Judging ran in batches of five pairs, with at least five seconds between attempts and up to three attempts per batch. Results were saved after each successful batch. HH needed five additional request attempts and UF needed four. Five persona comparisons still had invalid JSON after three attempts; retrying them individually with the same judge and rubric resolved all five. The seven existing judgments were retained, along with attempt logs and raw retry responses.

Persona model ($N = 12$)	Style / 5	Usefulness / 5	Judged safe
Unadapted SmolLM2 instruct	1.33	1.75	10/12
Persona LoRA	1.17	1.75	10/12

The adapter did not improve aggregate persona fidelity or usefulness. Safety scores were unchanged at 10/12 for both models.

6 Interpretation and error analysis

HH. Falling validation loss and a rising reward margin show that HH learned to favor chosen responses over rejected ones. The selected step-400 checkpoint reached 62.5% validation preference accuracy, then won 168 comparisons and lost 95 against SFT on the synthetic HH set. Some winning responses are clearer and shorter. HH explains satellite imagery more clearly on `hh-eval-42-0002`; on `hh-eval-42-0004`, both models refuse harassment, but HH is more concise. Safety is still inconsistent. HH gives harmful assistance on `hh-eval-42-0001`, and both models fail to refuse on `hh-eval-42-0000`. The 63.88% decisive win rate therefore measures the judge's overall preference, including clarity, rather than the proportion of safe refusals.

UF. UF reached validation DPO loss 0.585609 and preference accuracy 68.9%, better numerical values than HH's. Yet it lost 138 AlpacaEval comparisons against SFT and won 95. Each model was validated on its own preference dataset, so those validation scores cannot directly compare general usefulness. On `alpaca-42-0002`, SFT follows the sports-commentator instruction more closely while UF mixes incompatible sports details. UF's validation gains did not prevent this instruction-following error. It maintains character more consistently on `alpaca-42-0003`, while both models misinterpret an idiom on `alpaca-42-0000`. Overall, its 40.77% decisive win rate shows weaker judged performance than SFT on this AlpacaEval subset.

Persona. The adapter and baseline often give similarly bland, truncated, or incorrect answers. Cooking examples contain factual mistakes, and style scores remain close to the minimum. The answers to `persona-03` are identical. Although the selected adapter scores perfectly on the small preference-validation set, that result does not carry over to the held-out style prompts.

Both main DPO runs started from the same SFT checkpoint and used beta 0.1, the same learning rate, and a one-epoch budget. Decoding was matched within each comparison. Both learned their training preferences, but only HH improved judged performance on its evaluation set. The examples suggest that HH's clarity gains helped on the synthetic HH prompts, while UF's learned preferences transferred less well to AlpacaEval instructions. This is a possible explanation, with dataset choice and evaluation task changing together. A direct HH-versus-UF ranking would require testing both on the same set. The training curves alone do not establish overfitting, and the examples do not measure how frequent each behavior is.

7 Limitations and reproducibility

All comparisons use one judge, Gemini 3.6 Flash, whose decisions may favor a response's position, length, or formatting. Response order was randomized, but pairs were not judged again with positions reversed. Human assessment could therefore produce different results. Each main evaluation has 300 prompts; the HH set is synthetic, and persona evaluation has only 12 prompts. These sizes limit how broadly the results apply. Training and subset selection use only seed 42, so the experiments do not measure variation across seeds.

The experiments ran on Kaggle T4 ×2 with 512-token sequences, a 256-token prompt cap, and at most 256 generated tokens. These limits can remove context or cut answers short. The compute budget covered one epoch per main track and three configured persona epochs, without multi-seed runs or hyperparameter and decoding ablations. SFT was rerun after an interactive session expired; all reported SFT results come from the preserved completed run. Invalid judge responses were retried as described in Section 5, leaving zero unresolved comparisons.

Configurations are in `configs/`, with training metrics and plots in `results/training/`. Generated responses, pair mappings, and judgments are in `results/evaluation/`. Aggregate scores are saved in `results/evaluation-summary.json`, with attempt and failure records in `results/judging-history/`. The repository includes the fixed evaluation outputs and judgments, with hashes in `results/SHA256SUMS.txt`. For saved notebook versions and checkpoint locations, see `docs/ARTIFACTS.md`. Reusing the full weights requires access to the private saved Kaggle outputs.

References

[1] Rafailov et al. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. NeurIPS, 2023.

[2] Hu et al. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR, 2022.

[3] Course Staff. *Instruction Tuning and Direct Preference Optimization on a Small Language Model*, PA2 assignment manual.

[4] Hugging Face. TRL 0.15.2 DPOTrainer documentation. https://huggingface.co/docs/trl/v0.15.2/dpo_trainer.

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